

4.2 Network Security Services

In this course, I will:

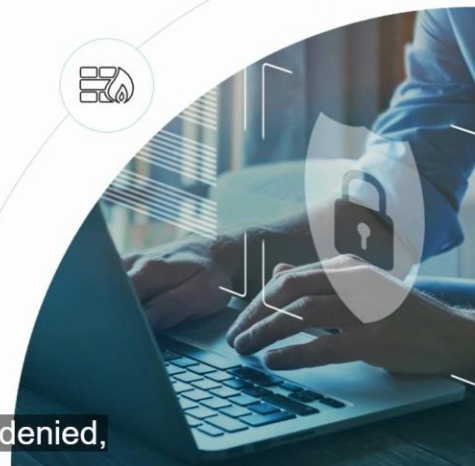
- Cover how to create network ACLs using the console, the CLI, and PowerShell
- Create rules in the network ACL to restrict in and outbound traffic, and associate the network ACL with a subnet
- Create security groups using the console, the CLI, and PowerShell, followed by creating security group rules and creating a virtual machine association
- Deploy the AWS Network Firewall for network perimeter protection including intrusion prevention

Network Access Control List (ACLs)

Network Access Control List (ACL)

- Packet filtering firewall solution
- Source/destination IP address
- Source/destination port
- Protocol type – TCP, UDP, ICMPv4, ICMPv6, custom

traffic should be allowed or denied,

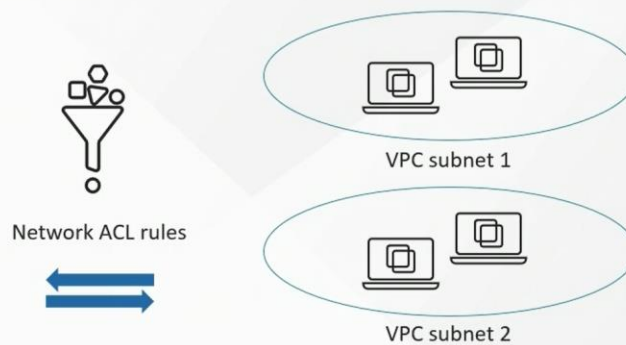


Network ACLs



Normally, the approach is to block everything with a network ACL

Network ACLs



we have two VPC subnets, subnet 1 and subnet 2.

Network ACLs



So, you can manage network ACLs using the AWS console, the CLI,

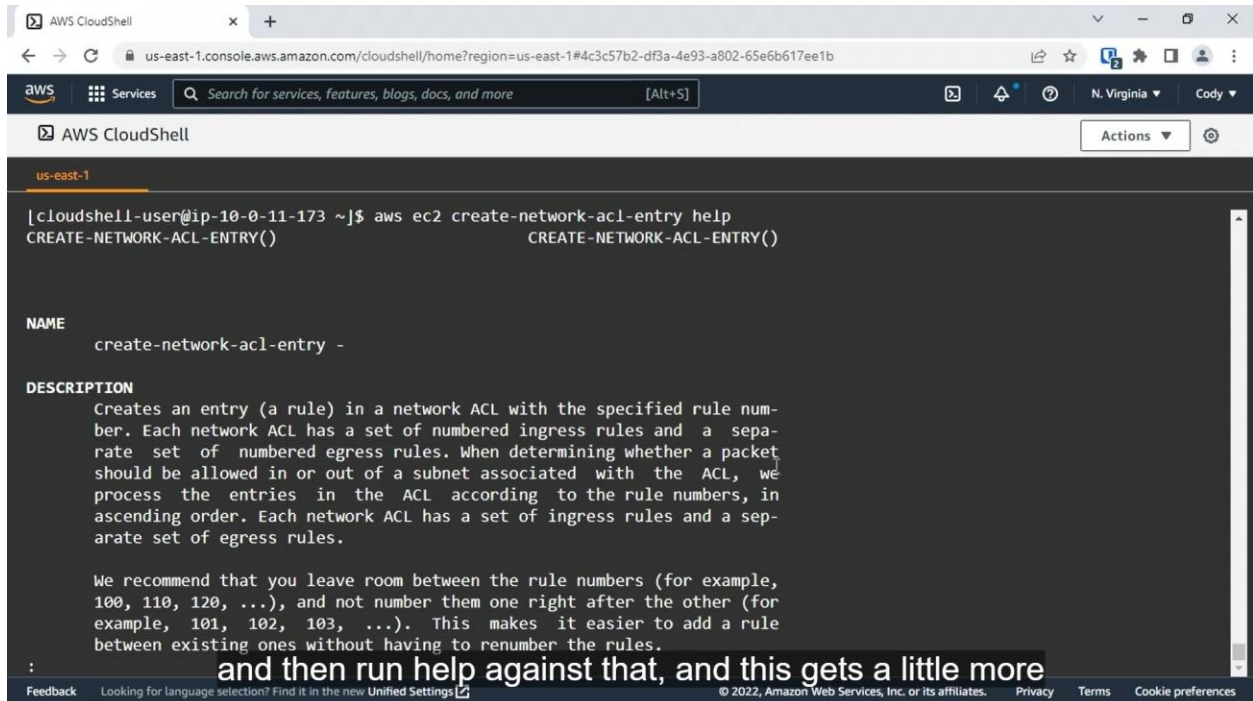
Configuring Network ACLs Using the AWS Console

Configuring Network ACLs Using the AWS CLI

The screenshot shows the AWS CloudShell interface. The terminal prompt is `[cloudshell-user@ip-10-0-11-173 ~]$`. The command `aws ec2 create-network-acl help` has been entered. The AWS console header is visible at the top, showing the AWS logo, a search bar, and the region set to N. Virginia.

```
[cloudshell-user@ip-10-0-11-173 ~]$ aws ec2 create-network-acl help
```

Therefore, I could run `aws ec2 create-network-acl help`,



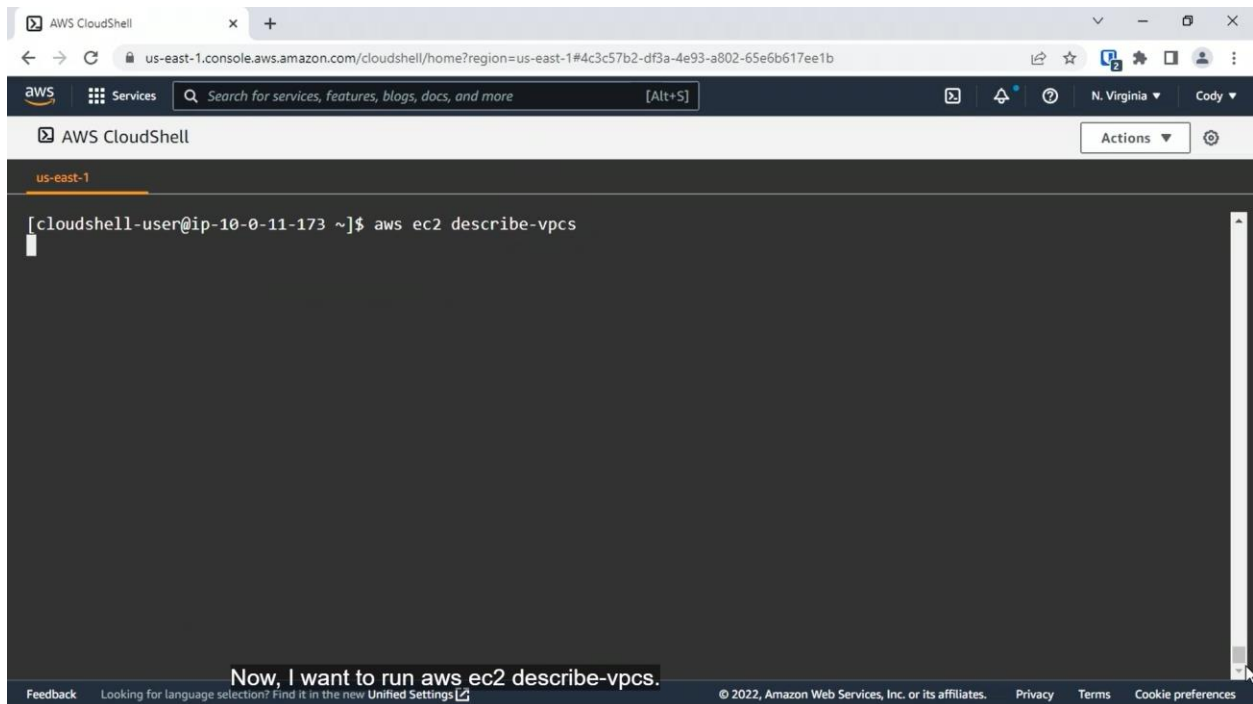
The screenshot shows the AWS CloudShell interface in a web browser. The terminal window displays the command `aws ec2 create-network-acl-entry help` and its output. The output includes the command name, a description of what the command does (creating an entry in a network ACL), and a recommendation to leave gaps between rule numbers. A text overlay at the bottom of the terminal reads: "and then run help against that, and this gets a little more".

```
[cloudshell-user@ip-10-0-11-173 ~]$ aws ec2 create-network-acl-entry help
CREATE-NETWORK-ACL-ENTRY()

NAME
    create-network-acl-entry -

DESCRIPTION
    Creates an entry (a rule) in a network ACL with the specified rule number. Each network ACL has a set of numbered ingress rules and a separate set of numbered egress rules. When determining whether a packet should be allowed in or out of a subnet associated with the ACL, we process the entries in the ACL according to the rule numbers, in ascending order. Each network ACL has a set of ingress rules and a separate set of egress rules.

    We recommend that you leave room between the rule numbers (for example, 100, 110, 120, ...), and not number them one right after the other (for example, 101, 102, 103, ...). This makes it easier to add a rule between existing ones without having to renumber the rules.
```



The screenshot shows the AWS CloudShell interface. The terminal window displays the command `aws ec2 describe-vpcs` which has been entered but not yet executed. A text overlay at the bottom of the terminal reads: "Now, I want to run aws ec2 describe-vpcs.".

```
[cloudshell-user@ip-10-0-11-173 ~]$ aws ec2 describe-vpcs
```

```
[cloudshell-user@ip-10-0-11-173 ~]$ aws ec2 create-tags --resources acl-0ce9249caf0868860 --tags Key=Name,Value=StandardWindowsNetworkACL
[cloudshell-user@ip-10-0-11-173 ~]$ aws ec2 create-network-acl-entry --network-acl-id acl-0ce9249caf0868860 --ingress --rule-number 100 --protocol tcp --port-range From=3389,To=3389 --cidr-block 0.0.0.0/0 --rule-action allow

[cloudshell-user@ip-10-0-11-173 ~]$ aws ec2 replace-network-acl-association --association-id aclassoc-bfe86ef3 --network-acl-id acl-0ce9249caf0868860
```

Configuring Network ACLs Using the PowerShell

```
PS C:\Users\Administrator> (get-ec2vpc).tags

Key                                     Value
---                                     -
Name                                   VPC1-vpc
aws:cloudformation:stack-name         EKSVPCStack
aws:cloudformation:stack-id           arn:aws:cloudformation:us-east-1:436376293329:stack/EKSVPCStack/7750cad0-14d0-11...
aws:cloudformation:logical-id         VPC
Name                                   EKSVPCStack-VPC
Name                                   default
Name                                   VPC2
aws:cloudformation:logical-id         Vpc
aws:cloudformation:stack-name         EC2ContainerService-cluster1
aws:cloudformation:stack-id           arn:aws:cloudformation:us-east-1:436376293329:stack/EC2ContainerService-cluster1...

PS C:\Users\Administrator> _
```

the cmdlet is not called (get-ec2vpc).tags.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> New-EC2NetworkAcl -VpcId vpc-009aee8072238d038

Associations : {}
Entries      : {Amazon.EC2.Model.NetworkAclEntry, Amazon.EC2.Model.NetworkAclEntry}
IsDefault    : False
NetworkAclId : ac1-02cdb81a3393fe5e0
OwnerId      : 436376293329
Tags         : {}
VpcId        : vpc-009aee8072238d038

PS C:\Users\Administrator> New-EC2Tag -ResourceId ac1-02cdb81a3393fe5e0 -Tag @{Key="Name"; Value="NetACL3"}
PS C:\Users\Administrator> _
```

OK, so I'm going to clear the screen.

```
Select Administrator: Windows PowerShell
PS C:\Users\Administrator> (get-ec2networkacl).tags

Key Value
---
Name AllowStandardLinuxPorts
Name StandardWindowsNetworkACL
Name NetACL2
Name NetACL3

PS C:\Users\Administrator> _
```


Security Groups

Security Groups

- Apply to EC2 instances, not subnets
- Have stateful firewall rules
- Control inbound traffic
- Control outbound traffic
- Can specify allow rules only
- Don't have a priority number



Also, security groups do not have priority numbers for individual entries,

Security Groups



You might opt where it makes sense to organize similar workloads into a

Security Groups

Can specify **allow** rules only – there is no deny rule option



Are **stateful** – changes to an incoming rule automatically apply for an outgoing rule



Have **all** their rules checked – there is no priority number



And all of the rules get checked within the security group.

Security Groups



Checked after network ACLs when both are in place

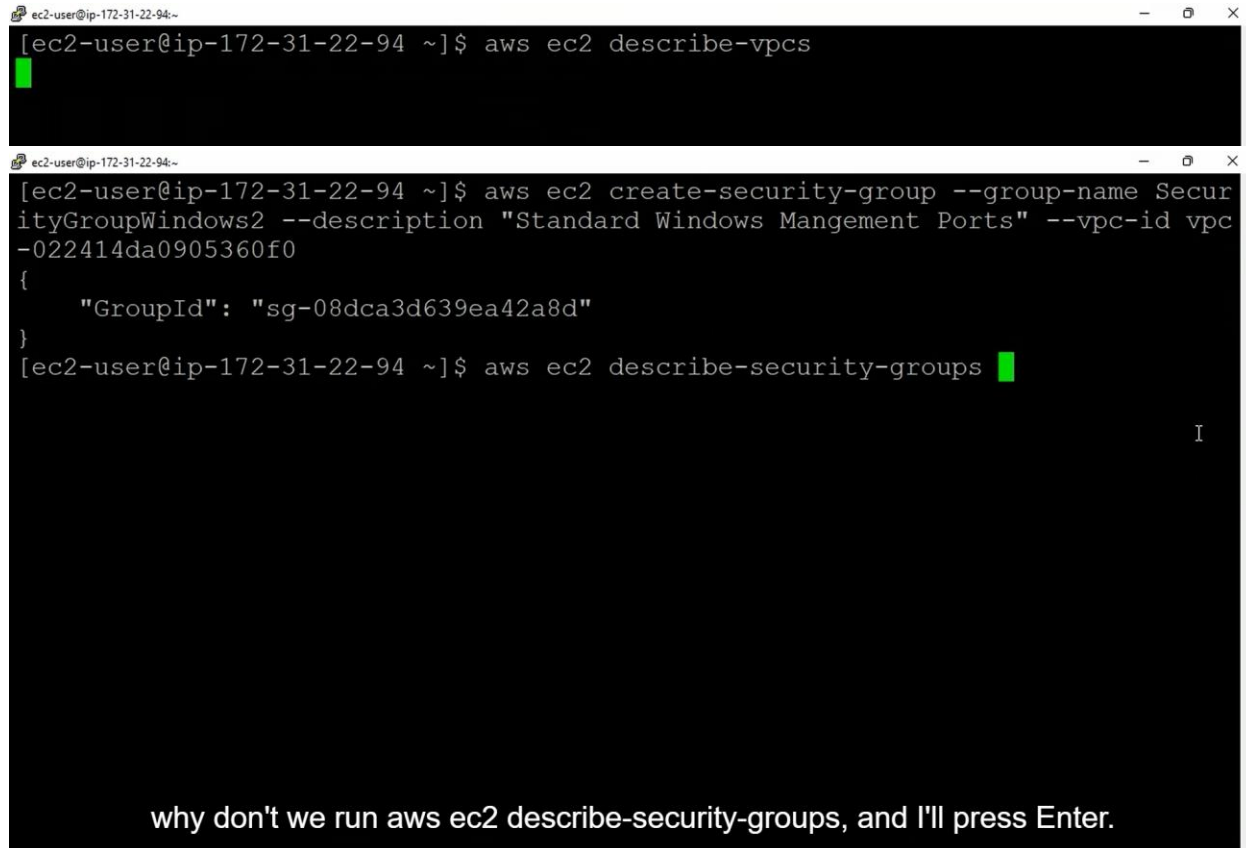


Managed using the console, CLI, PowerShell

So, we can manage security groups using console,

Configuring Security Groups Using the AWS Console

Configuring Security Groups Using the AWS CLI



```
ec2-user@ip-172-31-22-94:~  
[ec2-user@ip-172-31-22-94 ~]$ aws ec2 describe-vpcs  
  
ec2-user@ip-172-31-22-94:~  
[ec2-user@ip-172-31-22-94 ~]$ aws ec2 create-security-group --group-name SecurityGroupWindows2 --description "Standard Windows Mangement Ports" --vpc-id vpc-022414da0905360f0  
{  
  "GroupId": "sg-08dca3d639ea42a8d"  
}  
[ec2-user@ip-172-31-22-94 ~]$ aws ec2 describe-security-groups
```

why don't we run `aws ec2 describe-security-groups`, and I'll press Enter.

```

[ec2-user@ip-172-31-22-94 ~]$ aws ec2 describe-security-groups --query SecurityGroups[0].GroupName
"launch-wizard-43"
[ec2-user@ip-172-31-22-94 ~]$ aws ec2 describe-security-groups --query SecurityGroups[*].GroupName

```

```

[ec2-user@ip-172-31-22-94 ~]$ aws ec2 create-tags --resource sg-08dca3d639ea42a8d --tags Key=Name,Value="Security Group Windows 2"
[ec2-user@ip-172-31-22-94 ~]$ aws ec2 authorize-security-group-ingress --group-id sg-08dca3d639ea42a8d --protocol tcp --port 3389 --cidr 200.0.119.0/24
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-025b0182faf1f9d47",
      "GroupId": "sg-08dca3d639ea42a8d",
      "GroupOwnerId": "436376293329",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 3389,
      "ToPort": 3389,
      "CidrIpv4": "200.0.119.0/24"
    }
  ]
}
[ec2-user@ip-172-31-22-94 ~]$

```

OK, so it returns some output.

Configuring Security Groups Using PowerShell

```

Administrator: Windows PowerShell
PS C:\Users\Administrator> get-ec2vpc

```

```

Administrator: Windows PowerShell
PS C:\Users\Administrator> New-EC2SecurityGroup -GroupName "LinuxSecurityGroup2" -Description "LinuxSecGroup2" -VpcId vpc-009aee8072238d038
sg-0486283e11e0e68f2
PS C:\Users\Administrator> New-EC2Tag -ResourceId sg-0486283e11e0e68f2 -Tag @{Key="Name"; Value="Linux Security Group2"}
PS C:\Users\Administrator> $rule1 = @{IpProtocol="tcp"; FromPort="22"; ToPort="22"; IpRanges="199.126.129.0/24"}
PS C:\Users\Administrator> Grant-EC2SecurityGroupIngress -GroupID sg-0486283e11e0e68f2 -IpPermission $rule1

Return SecurityGroupRules
-----
True {sg-0486283e11e0e68f2}

PS C:\Users\Administrator>

```

Deploying AWS Network Firewall

Firewall

A firewall connects the VPC that you want to protect to the protection behavior that's defined in a firewall policy. For each Availability Zone where you want protection, you provide Network Firewall with a public subnet that's dedicated to the firewall endpoint.

To use the firewall, you update the VPC route tables to send incoming and outgoing traffic through the firewall endpoints.

Firewall Policy

A firewall policy defines the behavior of the firewall in a collection of stateless and stateful rule groups and other settings.

You can associate each firewall with only one firewall policy, but you can use a firewall policy for more than one firewall.

Rule group

A Network Firewall rule group is a collection of stateless or stateful rules that define how to inspect and handle network traffic.

Rules configuration includes 5-tuple and domain name filtering. You can also provide stateful rules using Suricata open source rule specification.

Firewall details

Name
Enter a unique name for the firewall.

the Firewall itself, Firewall Policy which can contain one or more Rule groups.

Your data is encrypted by default with a key that AWS owns and manages for you. To choose a different key, customize your encryption settings.

☐ Customize encryption settings (advanced)

Stateful rule group options

- ☒ **5-tuple**
Use the 5-tuple format, specifying the source IP, source port, destination IP, destination port, and protocol. Specify the action to take for matching traffic.
- ☐ **Domain list**
Specify a list of domain names and the action to take for traffic that tries to access one of the domains.
- ☐ **Suricata compatible IPS rules**
Intrusion prevention system (IPS) rules - Provide advanced firewall rules using Suricata rule syntax. Suricata is an open source network IPS that includes a standard rule-based language for traffic inspection.

Stateful rule order

The way that your stateful rules are ordered for evaluation.

Rule order

- ☒ **Default**
The stateful rules engine determines the evaluation order of your rules.
- ☐ **Strict**
You provide your rules in the order that you want them to be evaluated.

▼ **Rule variables - optional**

destination port number, and protocol type.

1. Which rule options are unavailable with security groups, but are available with Network ACLs?
 - Allow and deny actions

2. Which type of rule group inspects packets within the context of traffic flow?
- Stateful

3. What is wrong with the following PowerShell expression?

```
New-EC2NetworkAclEntry -NetworkAclId acl-0792ab793635523d7 -Egress $false -  
RuleNumber 100 -Protocol 6 -PortRange_From 443 -PortRange_To 443 -CidrBlock  
199.126.77.0/24
```

- The rule action was not specified

4. Which PowerShell cmdlet is used to create a security group rule?

- Grant-EC2SecurityGroupIngress

5. Which CLI command is used to create a Network ACL rule?

- aws ec2 create-network-acl-entry

6. Security groups can be associated with which type of AWS resources?

- EC2 instances

7. How is Network ACL rule processing determined?

- Rule number

8. You would like to create a security inbound rule allowing RDP management access. Which CLI syntax should you use?

- aws ec2 authorize-security-group-ingress

9. To which type of AWS resource are Network ACLs associated with?

- Subnet